

which hadoop

start-dfs.sh

start-yarn.sh

jps

cd ~/mapreduce

code . or go to the ubuntu home folder or search for mapreduce folder keep the mapper.py, reducer.py and test_input.txt files there

chmod +x mapper.py reducer.py – gives permission to run files to hdfs

hadoop fs -rm -r /output_result

hadoop fs -mkdir -p /input

hadoop fs -put -f test_input.txt /input/

hadoop fs -ls /input

find /usr/local/hadoop -name "hadoop-streaming*.jar"

2. How to make the command "Invincible"

If you are on a new PC and don't know where Hadoop is installed, you can search the **entire computer** by starting from the root directory (/):

Bash

```
find / -name "hadoop-streaming*.jar" 2>/dev/null
```

```
? /usr/local/hadoop
```

```
? /opt/hadoop
```

```
? /home/ubuntu/hadoop
```

COPY THE FIRST FILE PATH

```
hadoop jar [PASTE_JAR_PATH_HERE] \
```

```
-file mapper.py -mapper "python3 mapper.py" \
```

```
-file reducer.py -reducer "python3 reducer.py" \
```

```
-input /input/test_input.txt \
```

```
-output /output_result \
```

```
-cmdenv TARGET_WORD=apple
```

```
hadoop fs -cat /output_result/part-00000
```

```
hadoop fs -get /output_result/part-00000 ./final_count.txt
```

```
stop-all.sh
```

```
touch mapper.py – makes file
```

```
nano mapper.py – opens file
```

Delete the failed output folder:

```
hadoop fs -rm -r /output_result
```

EXTRAS

```
source ~/.bashrc
```

Always use `cd` to go into your folder first, and then use `pwd` to see exactly where you are.

? **HDFS Health:** <http://localhost:9870>

? **Job Tracker (YARN):** <http://localhost:8088>

Phase 1: Wake up the Cluster

First, make sure the Hadoop "engines" are running.

Bash

```
# Start storage services
```

```
start-dfs.sh
```

```
# Start resource management services
```

```
start-yarn.sh
```

```
# Verify everything is running (Look for NameNode, DataNode, etc.)
```

```
jps
```

Phase 2: Project Setup

Move into your folder and make sure your scripts are ready to go.

Bash

Go to your project folder

cd ~/mapreduce

Open VS Code to check your python code

code .

IMPORTANT: Give Ubuntu permission to run your scripts

chmod +x mapper.py reducer.py

Phase 3: The HDFS "Cleanup & Upload"

Hadoop will fail if the output folder already exists, so we always clean up first.

Bash

1. Delete old output from previous runs

hadoop fs -rm -r /output_result

2. Create input folder in Hadoop (if not already there)

hadoop fs -mkdir -p /input

3. Upload your text file from your computer to Hadoop

hadoop fs -put -f test_input.txt /input/

4. Verify the file is in Hadoop

hadoop fs -ls /input

Phase 4: Find the "Bridge" (Streaming JAR)

You need the exact path to the Streaming JAR to run Python.

Bash

```
find /usr/local/hadoop -name "hadoop-streaming*.jar"
```

Copy the first path that ends in .jar.

Phase 5: Execute the MapReduce Job

Paste the JAR path where indicated below. Note: The \ allows you to write one command over multiple lines for better readability.

Bash

```
hadoop jar [PASTE_JAR_PATH_HERE] \  
-file mapper.py -mapper "python3 mapper.py" \  
-file reducer.py -reducer "python3 reducer.py" \  
-input /input/test_input.txt \  
-output /output_result \  
-cmdenv TARGET_WORD=apple
```

Phase 6: Show the Results

Once the terminal says "Job completed successfully," show your work.

To show the result in the terminal:

Bash

```
hadoop fs -cat /output_result/part-00000
```

To download the file to your local folder to show the examiner:

Bash

```
hadoop fs -get /output_result/part-00000 ./final_count.txt
```



Quick Troubleshooting for the Exam:

- **If jps shows nothing:** Your Hadoop variables aren't loaded. Run `source ~/.bashrc`.
- **If it says "Permission Denied":** You forgot `chmod +x mapper.py reducer.py`.

- **If it says "Output directory already exists":** Run `hadoop fs -rm -r /output_result`.
- **If the count is 0:** Check if the word in `-cmdenv TARGET_WORD=...` actually exists in your `test_input.txt`.

Good luck with your exam

Mapper.py

```
#!/usr/bin/env python3
```

```
import sys
```

```
# The target word we are looking for (passed as an environment variable)
```

```
import os
```

```
target_word = os.environ.get('TARGET_WORD', 'hadoop').lower()
```

```
for line in sys.stdin:
```

```
    # Clean and split the line
```

```
    words = line.strip().split()
```

```
    for word in words:
```

```
        # Remove punctuation and lowercase
```

```
        clean_word = "".join(filter(str.isalnum, word)).lower()
```

```
        if clean_word == target_word:
```

```
            print(f"{clean_word}\t1")
```

reducer.py

```
#!/usr/bin/env python3
```

```
import sys

current_word = None
current_count = 0

for line in sys.stdin:
    word, count = line.strip().split('\t', 1)
    count = int(count)

    if current_word == word:
        current_count += count
    else:
        if current_word:
            print(f"{current_word}\t{current_count}")
        current_word = word
        current_count = count

if current_word:
    print(f"{current_word}\t{current_count}")
```

Input.txt

apple banana apple cherry